A blockchain based system runs on the concept of human trust. A blockchain network is built in such a way that it presumes any individual node could attack it at any time. The consensus protocol, like proof of work, ensures that even if this happens, the network completes its work as intended, regardless of human cheating or intervention. The blockchain allows one to secure stored data using various cryptographic properties such as digital signatures and hashing. As soon as the data enters a block in the blockchain, it cannot be tampered with and this property is called immutability. If anyone tries to tamper with the blockchain database, then the network consensus will recognise the fact and shut down the attempt.

Blockchains are made up of nodes; these can be within one institution like a hospital, or can be all over the world on the computer of any citizen who wants to participate in the blockchain. For any decision to be made, the majority of the nodes need to come to a consensus. The blockchain has a democratic system instead of a central authoritarian figure. So if any one node is compromised due to malicious action, the rest of the nodes recognise the problem and do not execute the unacceptable activity. Though blockchain has a pretty incredible security feature, it is not used by everyone to store data.

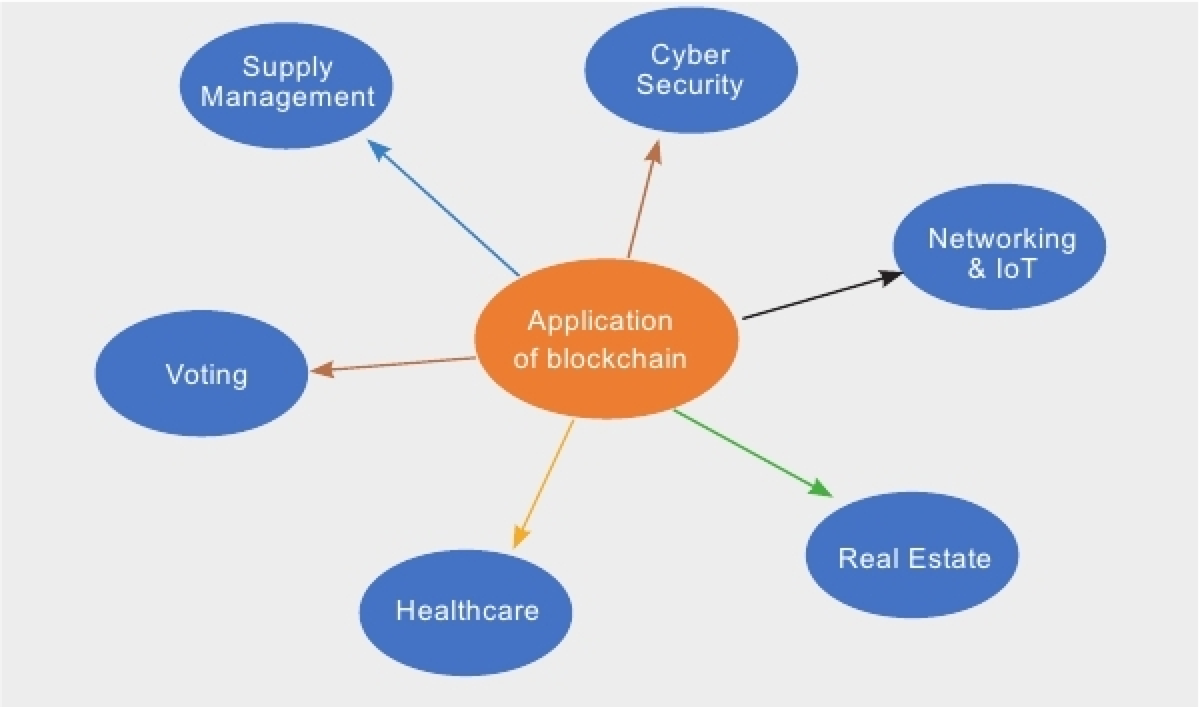

Figure1: Applications of blockchain

Common use cases of blockchain in cyber security

Mitigating DDoS attacks: A distributed denial-of-service attack is a cyber-attack; it involves multiple compromised computer systems that aim at a target and attack it, causing denial of service for users of the targeted resources. This causes the system to slow down and crash, hence denying services to legitimate users. There are some forms of DDoS software that are causing huge problems. One among them is Hide and Seek malware, which has the ability to act even after the system reboots and hence can cause the system to crash over and over again. Currently, the difficulty in handling DDoS attacks is due to the existing DNS (Domain Name System). A solution to this is the implementation of blockchain technology. It will decentralise the DNS, distributing the data to a greater number of nodes and making it impossible for the hackers to hack.

More secure DNS: For hackers, DNS is an easy target. Hence DNS service providers like Twitter, PayPal, etc, can be brought down. Adding the blockchain to the DNS will enhance the security, because that one single target which can be compromised is removed.

Advanced confidentiality and data integrity: Initially, blockchain had no particular access controls. But as it evolved, more confidentiality and access controls were added, ensuring that data as a whole or in part was not accessible to any wrong person or organisation. Private keys are generally used for signing documents and other purposes. Since these keys can be tampered with, they need to be protected. Blockchain replaces such secret keys with transparency.

Improved PKI: PKI or Public Key Infrastructure is one of the most popular forms of public key cryptography which keeps the messaging apps, websites, emails and other forms of communications secure. The main issue with this cryptography is that most PKI implementations depend on trusted third party Certificate Authorities (CA). But these certificate authorities can easily be compromised by hackers and spoof user identities. When keys are published on the blockchain, the risk of false key generation is eliminated. Along with that, blockchain enables applications to verify the identity of the person you are communicating with. ‘Certain’ is the first implementation of blockchain based PKI.

The major roles of blockchain in cyber security

Eliminating the human factor from authentication: Human intervention is eliminated from the process of authentication. With the help of blockchain technology, businesses are able to authenticate devices and users without the need for a password. Hence, blockchain avoids being a potential attack vector.

Decentralised storage: Blockchain users’ data can be maintained on their computers in their network. This ensures that the chain won’t collapse. If someone other than the owner of a component of data (say, an attacker) attempts to destroy a block, the entire system checks each and every data block to identify the one that